

ECEN 743: Reinforcement Learning

Assignment 4

Overview

1. You will complete this assignment in a Jupyter notebook. You can download it from our (ECEN743 GitHub repo).
2. You have to submit an HTML report, your code and two videos to Canvas.
3. Put all your files (HTML report, code and videos) into a **single compressed folder** named **LastName_FirstName_A3.zip**.
4. Refer to the README of the GitHub repository for how to generate the proper HTML report for submission and for the installation instructions.

In this homework, you will train a deep Q-Learning algorithm to land a lunar lander on the surface of the moon. We will use the Lunar Lander environment (LunarLander-v3) from Gymnasium. The environment consists of a lander with 4 discrete actions and a continuous state space. A detailed description of the environment can be found [here](#). The problem is solved if the total reward per **episode is 200 or above**. Do not stop training on the first instance your agent gets a reward above 200, your agent must achieve a reward of 200 or above consistently.

1. Deep Q-Learning

Implement deep Q-learning **with** experience replay and target network. You should include the training curve in the HTML report (x-axis is the number of episodes and the y-axis should be the episodic cumulative reward). Use a sliding window average to get smooth plots. Include a description of the hyperparameters used. Also, you should submit a video of the smooth landing achieved by your RL algorithm. Try to find the optimal hyperparameters that will enable fast convergence to the optimal policy.

2. Ablation Study

Perform an ablation study to understand the importance of experience replay and target network. You should include the training curves, and describe your observations and inferences.

3. Algorithmic Improvements

As we discussed in class, there have been a number of improvements in the deep Q-learning algorithm and architecture, such as double DQN, dueling DQN, prioritized experience replay, and many others (see the slides and related papers). Implement double DQN. Explain your implementation and demonstrate the benefit.

4. Deep Q-Learning in Another Environment

Congratulations on implementing deep Q-Learning on the Lunar-Lander-v3 environment! Now, learn the optimal policy for another control problem (environment) using deep Q-learning. You can select one environment from the *Classical Control* set, *Box2D* set, or *Atari Games* set in Gymnasium. In the HTML report, you need to clearly specify the environment and provide a link to the corresponding page in Gymnasium. You need to include the training curve and describe the hyperparameters used. You should also include the video of the performance.